Technical Assignment – Senior Frontend Engineer

Task Overview
Build a diagram editor for visualizing and managing large node networks using React, TypeScript, GoJS, MUI, and Jest.

Product Goals
• Visualize large and potentially growing node networks
• Smooth interaction while browsing and editing
• Quick node management via a side panel
• Easy editing: add nodes, create links, update properties
• Instant sync: selection and edits update everywhere
• Scalable and maintainable implementation

Requirements

Data Model
Node = { id: string; name: string; type: 'Node' }

Diagram (GoJS)
• Render a graph with 1000 connected nodes
• Support selection
• Allow adding nodes and creating links
• Avoid unnecessary re-renders or expensive operations
• Must not re-initialize the diagram on every render

Side Panel (MUI)
• Node list showing all nodes
• Efficient rendering (no external virtualization libraries)
• Selection sync with the diagram (both directions)
• Properties panel: edit node name, show node type (read-only)

Synchronization
Diagram ↔ React state ↔ Diagram
No duplicated or conflicting state

State Management
Use React state only (no external state management libraries)

Testing
Provide tests for core functionality:
• Node generation
• Adding a node
• Linking nodes
• Updating a node name
Focus on meaningful scenarios, not only happy paths

Additional Notes
• Provide a GitHub repository or ZIP
• Include a README describing your approach, structure, and decisions
• If AI tools were used:
- What parts were generated
- What you modified or rejected
- What you verified manually